

Implicit and Explicit Visual Framing Effects on Illusory Relative Weight Perception in Virtual Reality

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ABSTRACT

In virtual reality, weight perception can be framed through visual cues only, yet effects of different types of visual framings has not been fully explored. We experimentally investigate the individual/interactive effects of implicit (by visual surface) and explicit framing (overt depiction of deformation) and when combined in a congruent or possibly conflicting way. Preliminary results demonstrated that explicit framing dominated the implicit. Correct selection rates peaked when framings aligned congruently. These findings characterize how explicit and implicit visual framings can bias illusory relative weight judgments in VR and provide insights into how to apply them in conjunction with pseudo-haptic techniques.

Index Terms: Virtual Reality, Weight Perception, Visual Framing, Implicit Cues, Explicit Cues

1 INTRODUCTION

Virtual reality environments enable manipulation of visual-only feedback to bias perceptual physical experiences without haptic devices. Prior work shows that purely visual cues such as material appearance and dynamic deformation can induce illusory changes in relative weight perception [2, 1], yet these studies typically examine a single framing type in isolation. How users integrate multiple, potentially conflicting visual weight framings during real-time interaction remains largely unexplored.

Visual weight framings operate through distinct cognitive mechanisms. Implicit framing relies on learned material associations and semantic memory, whereas explicit framing provides immediate perceptual evidence via dynamic visual feedback. Characterizing which type dominates is critical for designing pseudo-haptic VR systems that enhance perceived physicality without kinesthetic devices.

2 EXPERIMENT

Design: Participants completed a within-subjects 2x2 factorial experiment that combined implicit and explicit visual framing. Implicit framing was defined as the static material appearance of the cubes (metal versus wood), relying on learned real-world associations. Explicit framing was defined as the degree of cushion deformation, together with an instruction that “more deformation means heavier.” These two framings yielded four conditions (implicit-only, explicit-only, congruent, incongruent), each tested with a two-alternative forced-choice task with 25 trials per condition (100 trials in total). Four tested conditions given in balanced order were as follows (see Figure 1):

- IMP (implicit): cubes shared identical physical simulation with no haptic difference but differed in rendered material.

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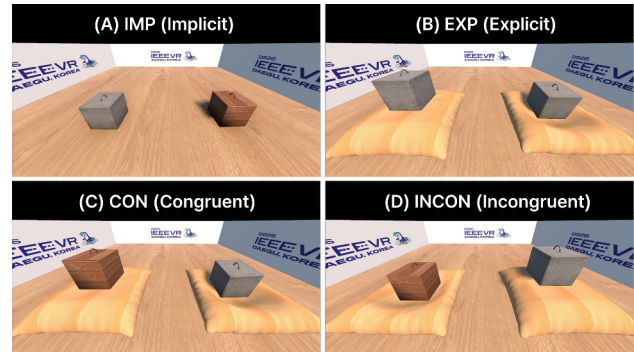


Figure 1: Four experimental framing methods comprising implicit, explicit, and combined framing conditions.

- EXP (explicit): same metal material with a fixed deformation depth difference between the two cubes on every trial, visually suggesting weight differences without changing physical mass.
- CON (explicit and implicit congruent): material and deformation aligned (metal+strong deformation vs. wood+weak deformation).
- INCON (explicit and implicit incongruent): framings conflicted (metal+weak deformation vs. wood+strong deformation), serving as the critical dominance assessment.

Participants: 16 adults (10M/6F, age 31.4±4.9, all with VR experience) participated in the experiment.

Apparatus: We conducted the experiments using Meta Quest 3 HMDs and Unity-based software.

Procedure: The order of the experimental conditions was counterbalanced using a 4x4 Latin Square design to control for order effects. Each trial presented two cubes on a table. Participants sequentially lifted each cube via hand tracking, then selected the perceived heavier object via gaze-based button (see supplementary video).

Measures: Following dependent variables were measured: (1) selection rate, defined as the proportion of trials choosing the cube visually framed as heavier (uniquely determined by the manipulated cue(s) in IMP, EXP, and CON, and defined as the cube with stronger cushion deformation in INCON), (2) reaction time (RT) per trial, (3) Visual Dependence Scale (VDS, 5 items) scores adapted from prior VR spatial orientation research [4], (4) subjective difficulty ratings (1–5), and (5) self-reported decision strategy (material appearance, cushion deformation, both equally, or intuition).

3 RESULTS

3.1 Selection Patterns

All conditions produced selection rates significantly above the 50% indifference point (all $p < 0.001$). The results showed as

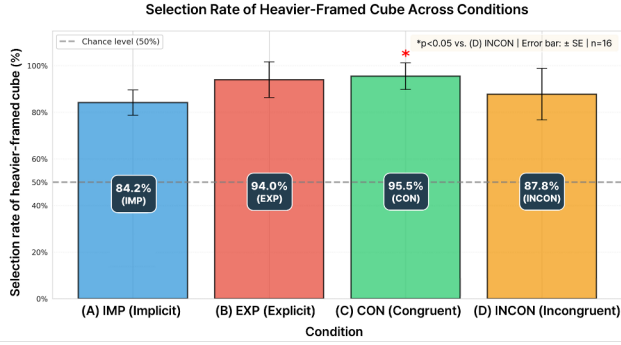


Figure 2: Selection rates across conditions. All conditions significantly exceed 50% indifference point.

follows: IMP: the metal cube was rated $84.2\% \pm 13.3\%$ heavier than the wooden cube due to implicit material property framing. ($d = 2.570$); EXP: large deformation under explicit weight framing resulted in a $94.0\% \pm 18.8\%$ rate of heavy object selection ($d = 2.341$); CON: A higher selection rate of $95.5\% \pm 13.9\%$ ($d = 3.270$) was observed when implicit and explicit visual framing were congruent. Additionally, repeated-measures ANOVA revealed no significant differences among the three conditions ($F(2, 30) = 1.18, p = 0.321$), suggesting that all framing types exerted a comparable influence on biasing user choices.

As for INCON, where the “heavier-framed” cube was defined by the explicit deformation cue, explicit framing dominated with a mean selection rate of 87.8% selection rate ($SD=27.1\%$, $t(15) = 5.580, p < 0.001, d = 1.395$). Individual analysis showed 13 participants (81%; out of 16) exhibited strong deformation-framing preference ($\geq 90\%$), while only 1 participant preferred material framing. Self-reported strategy aligned with these results - participants reporting preference for deformation-framing showed 95–100% deformation selection ($r = 0.76$).

3.2 Cognitive Processing

Reaction times differed significantly across conditions; IMP: $8.50 \pm 5.66s$; EXP: $4.02 \pm 3.54s$; CON: $2.40 \pm 1.34s$; INCON: $6.80 \pm 4.31s$. Post-hoc Wilcoxon tests confirmed CON was the fastest (all $p < 0.05$), consistent with parallel processing of congruent cues. IMP required longest processing, reflecting semantic memory retrieval. INCON showed moderate RT increase, suggesting efficient conflict resolution through explicit-framing prioritization.

3.3 Subjective Metrics

Visual Dependence Scale scores ($M=20.3 \pm 2.2$) showed no correlation with INCON selection ($r = 0.135, p = 0.614$), indicating illusory relative weight perception operates independently of general visuo-spatial processing. Subjective difficulty ratings correlated with IMP selection rate ($r = -0.672, p = 0.004$), validating meta-cognitive awareness. Self-reported strategies revealed: 69% “deformation-first”, 19% “both cues”, 12% “intuition (others)”.

4 DISCUSSION

Explicit framing overwhelmingly dominated implicit framing, with 87.8% of selecting deformation-framed cubes during conflicts. This aligns with multi-sensory integration theory [3], which posits that sensory systems optimally weight information sources by perceived reliability. In haptic-free VR, explicit deformation framing provides immediate visual evidence through real-time deformation,

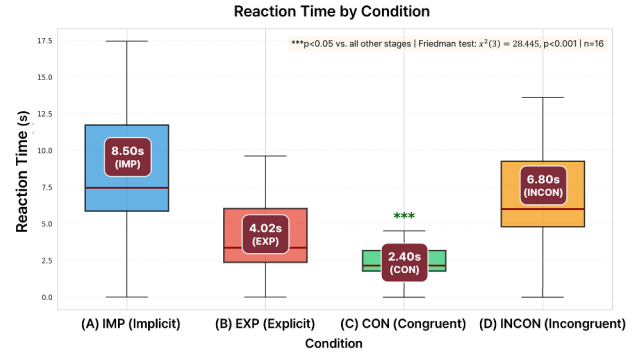


Figure 3: Reaction time distributions across conditions.

yielding higher reliability than implicit material framing, which requires retrieving learned associations [2]. This explains systematic prioritization of overt cues under conflict.

Design Guidelines: Based on the findings, we propose the following design guidelines: (1) leverage material appearance for initial expectations, (2) prioritize dynamic deformation for perceptual clarity, and (3) strategically use conflicting framings to override assumptions or create novel weight experiences. These principles apply to pseudo-haptic interfaces, VR training, and immersive applications.

Future Work: Future work will explore alternative explicit modalities for weight modulation and assess the integration of these pseudo-weight illusions into complex, high-fidelity VR scenarios to enhance real-world applicability. Furthermore, investigating diverse deformation patterns—such as squishing or bending—will be crucial to validating the generalizability of our findings across various dynamic framing types.

5 CONCLUSION

This study demonstrates that visual modulations can effectively alter weight perception in VR without physical haptic changes. While implicit cues derived from prior knowledge influence perception, explicit visual deformation provides a significantly more robust modulation of perceived weight. Notably, when these implicit and explicit cues are congruent, users exhibit superior selection consistency and faster weight processing speeds. These findings establish evidence-based principles for designing illusory weight systems. Ultimately, these insights can be readily integrated into VR applications to enhance interactive realism.

REFERENCES

- [1] Y. Ban, T. Narumi, T. Tanikawa, and M. Hirose. Modifying an object’s trajectory affects the perception of its weight through integral bias. *IEEE Transactions on Visualization and Computer Graphics*, 18(12):2049–2057, 2012. 1
- [2] G. Buckingham, M. A. Goodale, and J. R. Flanagan. The material-weight illusion is determined by visual experience and grasp execution. *Neuropsychologia*, 47(3):707–714, 2009. 1, 2
- [3] M. O. Ernst and M. S. Banks. Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*, 415(6870):429–433, 2002. 2
- [4] A. Maneuvrier, L. M. Decker, H. Ceyte, P. Fleury, and P. Renaud. Field (in)dependence flexibility following a virtual immersion in a 3d environment and its subjective experience. In *Frontiers in Virtual Reality*, vol. 2, p. 706712. Frontiers, 2021. 1